BetWise – Technical documentation

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This is the technical documentation of BetWise, a prototype of medical device developed during the "Laboratory of Medical Devices and Systems" of the Bachelor of Science in Artificial Intelligence by the students:

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Device description and specification

General description of the AI-based device

BetWise is an AI-based medical device conceived as a plugin for the IO App. It has been designed to analyze users' gambling activity and to identify early signs of addiction. By evaluating behavioral markers such as frequency, intensity, variability and trajectory of gambling activity, the system generates predictions and classifies users into one out of four distinct risk levels.
Based on this classification, the device delivers tailored feedback and, when deemed appropriate, recommends the users to consult a medical professional for further evaluation which could, eventually, lead to a formal diagnosis of gambling addiction.

Intended purpose

BetWise is conceived to detect early signs of addiction by analysing users' self-reported gambling activity collected though a questionnaire directly accessible through the IO App (https://ioapp.it/). The data collected allows a centralized system to classify users into one out of four classes, corresponding to different risk levels of addiction development, and returning a recommendation whether to seek for medical consultation.

The device is designed to detect patterns associated with the development of gambling addiction, but it does not diagnose or treat this clinical disorder. Its primary goal is to raise awareness and to promote responsible behavior, functioning as a supportive tool that encourages users to seek for appropriate care, when necessary.
There are no known medical contraindications to the use of BetWise; however it is important to highlight that BetWise is not a diagnostic tool and should not be used as a substitute for professional medical evaluation. In fact, all the predictions emitted by our medical device must be reviewed and interpreted by qualified medical professionals.

The system has been conceived for adult Italian citizens, users of the IO app, especially those involved in (online) gambling activities, although it could be adopted more broadly by every adult as a form of prevention and detection of potentially harmful gambling behaviors. The tool should be directly integrated into the IO app, ensuring easy access and aiming at reaching a wider crowd as possible, other than automatically validating the quality and origin of data.

AI interaction

Our medical device is integrated as a plugin within the IO App, prompting users to complete a brief questionnaire regarding their online betting behavior. The data collected through this questionnaire is transmitted in real-time to a secure cloud-based AI module, where it is analyzed using an XGBoost classifier.
Based on the provided responses, the AI system classifies users into one of four behavioral categories, each associated with a different risk level:

- **Class 1** (*high intensity, high variability*), gamblers who play frequently and intensively, with high variability in wager size. It represents the highest risk profile.

- **Class 2** (*low first month activity*), gamblers who play very rarely. Despite the low activity, gambling-related issues can be identified.
- **Class 3** (*high intensity, low variability*), gamblers who play as frequently and intensively as members of Class 1, but with stable and consistent wager sizes. The absence of variability suggests a more controlled behaviour and, consequently, a lower risk of addiction development associated with the members of this class.
- **Class 4** (*moderate betting*), the majority of gamblers, who play infrequently and non-intensively, with low variability in wager size. This profile is associated with the lowest likelihood of developing gambling-related problems.

The classification output is returned to the user directly via the IO app, along with tailored advice. In high risk scenarios, the system recommends considering a medical consultation, which may lead to a formal diagnosis of Gambling Addiction (GA).

Technical specification

The device consists of a software-based AI system, which operates without a dedicated firmware and is entirely cloud-based. Its main component is a pre-trained XGBoost classifier which processes users' responses to a questionnaire in order to classify them into one of the four risk categories previously defined.
The system processes the data on a centralized server, while, on the client side, the device appears as a simple plugin embedded in the IO app. Users interact with the system through the IO app interface, which facilitates the collection of data and displays the resulting classification along with any related recommendation.

The backend is developed in Python, supported by the following libraries and versions:

• xgboost	version 3.0.0
• scikit-learn	version 1.6.1
• pandas	version 2.2.3
• numpy	version 2.2.0
• seaborn	version 0.13.2
• matplotlib	version 3.10.3
• scipy	version 1.15.2
• imbalanced-learn	version 0.13.0
• mlxtend	version 0.23.4
• missingno	version 0.5.2
• SHAP	version 0.47.1

All software dependencies, which are maintained at their latest stable release, are handled server-side and do not require any user intervention.
Since the processing is centralized, there isn't any particular hardware requirement and the system operates within the minimum standard specifications defined for the IO app. In particular, the IO app is supported on Android devices running version 7.1 or higher and on iPhones with iOS 14 or later. These are the only prerequisites for accessing to BetWise.

The user interface (UI) allows individuals to input responses and receive personalized output consisting of a risk level and, in high risk cases, also of a recommendation to consult a healthcare professional.

For deployers, the device requires no technical setup beyond standard plugin deployment within the IO App, as well as no additional configuration or manual oversight after integration.

Designation/Classification

BetWise qualifies as a medical device software (MDSW) according to the EU's Medical Devices Regulation (MDR) 2017/745. According to Article 2 of the MDR, a medical device is defined as any instrument, apparatus, appliance, software, implant, reagent, material, or other article intended by the manufacturer to be used, alone or in combination, for human beings for one or more of the specific medical purposes of diagnosis, prevention, monitoring, prediction, prognosis or treatment of disease and which does not achieve its principal intended action by pharmacological, immunological or metabolic means.

In accordance with this definition, BetWise can be defined as a medical device as it is a software-based tool intended to be used for the prevention, monitoring and early prediction of gambling addiction. These scopes are achieved by analysing behavioral patterns and providing risk classifications along with recommendations, if needed, to the (adult) users of the IO app.

This system is not an in vitro diagnostic device and, therefore, does not fall under the scope of Regulation (EU) 2017/746 (IVDR). In fact, as defined in article 2 of the IVDR, a medical device in order to be considered an in vitro diagnostic medical device should be intended for the examination of specimens derived from the human body. BetWise, however, doesn't analyze biological specimens, but operates solely on self-reported behavioral data, processed with AI.

Following the MDR Annex VIII, Rule 11, which is specific to MDSW: “Software intended to provide information which is used to take decisions with diagnosis or therapeutic purposes is classified as Class IIa, unless such decisions have an impact that may cause: (a) death or an irreversible deterioration of a person's state of health, in which case it is in Class III; or (b) a serious deterioration of a person's state of health or a surgical intervention, in which case it is classified as Class IIb.”
BetWise is classified under the low **risk class IIa**, as it provides predictive and monitoring information that can influence clinical decision-making related to addiction risk, but does not directly control therapy nor pose immediate or irreversible health risks.

Furthermore, under the AI Act, BetWise qualifies as a high-risk AI system, given its classification as a medical device under the MDR. Consequently, the system must comply with the applicable high-risk AI obligations, including conformity assessment, transparency, post-market monitoring and risk management procedures.

Design and manufacturing information

Description of the design

The strategy of integrating BetWise as a plugin of the IO App guarantees both the authenticity and the integrity of the data and, in addition to the choice of close-ended questions for the poll, it ensures operability and accessibility to the tool.

The core idea is to provide a simple questionnaire, easy to complete yet effective in assessing and classifying the severity of addiction risk into different levels. This approach is not meant to replace a medical diagnosis, but rather to serve as an early warning step that could help citizens recognize whether they may be at risk and, therefore, need to consult a medical practitioner. These principles are what drove the AI development described below.

Description of the AI development

As mentioned above, the aim of the prototype is to provide useful insights about people's tendency to develop gambling addiction (GA), providing them with a suggestion to consider (if estimated as necessary) a medical consultation which could lead to a GA diagnosis. In essence, the model tries to predict the medical diagnosis by classifying the user into one of 4 classes representing different levels of hazardous behavior and, according to its findings, it suggests to the user whether a medical consultation might be helpful.

Online gambling activity has been deeply studied before by **LaBrie et al**. in a prospective longitudinal study of actual Internet sports gambling behavior during eight months (LaBrie et al. Assessing the Playing Field: A Prospective Longitudinal Study of Internet Sports Gambling Behavior. J Gambl Stud 23, 347–362 (2007). https://doi.org/10.1007/s10899-007-9067-3); they "tracked the following primary gambling behaviors: daily totals of the number of bets made, money bet, and money won" and they transformed these variables into measures of gambling involvement.

More recently, **Braverman et al.** (Braverman J, Shaffer HJ. How do gamblers start gambling: identifying behavioural markers for high-risk internet gambling. Eur J Public Health. 2012 Apr; 22(2):273-8 https://doi.org/10.1093/eurpub/ckp232) proposed a clustering analysis aimed at providing "important information that can help to identify potentially problematic gamblers during the early stages of gambling-related problems" by identifying betting patterns during the first month of online activity.

The authors identified 4 clusters:

- **Cluster 1** (high intensity, high variability) contained gamblers who played frequently and intensively, with high variability of the wager sizes and positive trajectory.
- **Cluster 2** (low first month activity) contained gamblers who played very rarely.
- **Cluster 3** (high intensity, low variability) consisted of gamblers who played as frequently as the members of Cluster 1 and intensively, but bet about the same amount of money each day that they gambled with almost no increase in the waging size.

- **Cluster 4** (moderate betting) included the majority of gamblers: those who played rarely, not intensively, with low variability in their wager size and the trajectory close to zero.

Gamblers characterized by high-intensity and frequency of gambling and by high variability of wager sizes were at higher risk to report gambling-related problems.

We wanted to train a classifier able to associate one's gambling behavior to the class of hazardousness (corresponding to one of the clusters found by Braverman et al.). In order to accomplish that, we developed a machine-learning (ML) pipeline based exactly on the same data adopted by the researchers: numerically collected tabular data aimed at reporting variability, frequency and intensity of betting.

The dataset reported no missing value and it was composed of a total of 18 features (plus the Ground Truth representing the cluster of each user). According to our needs, we kept only 14 of them, 2 of those were categorical ('Sereason', one out of three reasons for the closure of the online account; 'Gender', user's binary gender); all the remaining features were purely numerical.
Both the categorical features and the labels were discretely/highly unbalanced; in particular, males were many more than females and class 4 (inherited by cluster 4) was the most recurrent, while class 1 and 2 were much lower represented.
Numerical features were either normally distributed or skewed; therefore, we proposed a preprocessing pipeline able to handle both cases.

We initialized a Perceptron classifier (before choosing the optimal one) with PCA and SMOTE oversampler in order to handle class imbalance and to try to reduce the number of features. As shown in the Feature Correlation Matrix, total number of stakes, st. dev. of stakes during the first month, number of bets per day during the first month and total number of active days during the first month were the most related (and, therefore, useful) to the prediction.

We, then, proposed a nested cross validation adopting an iterative loop to search among 60 possible configurations of oversampler, dimensionality reduction strategy, classifier and hyperparameters. Our search space was composed as follows:

- Sampler: None, SMOTE, Random OverSampler;
- Dimensionality reduction: None, PCA, LDA, Sequential (forward) Feature Selection (SFS);
- Classifier: Perceptron, Logistic Regression, KNN, Random Forest, XGB; each of those with different hyperparameters' configurations to be tested.

We selected the 5 best performing configurations; we retrained them and we assessed their performance on novel data. All ensemble-learning methods tended to overfit; we prioritized a model with higher performance on novel data: XGB with no oversampling and no dimensionality reduction.

We ran hyperparameter optimization on the chosen configuration; in particular, we tested different numbers of estimators (10, 50, 100, 500), different learning rates (0.001, 0.01, 0.1, 0.3, 0.5) and different depths (4, 6, 8, 10). We found the optimal performance wrt F1 score by

using XGBClassifier(learning_rate=0.1, max_depth=8, n_estimators=50), reaching almost 93%.

As anticipated, the learning curve showed data overfitting, but the classifier reached a great stability on the test data as the training examples increased. The validation curve, indeed, confirmed this convergence: by correctly tuning the hyperparameters, F1 on the test set becomes stable and tends to converge to the score computed onto the training data.

Fairness and explainability analyses have been performed using SHAP. We displayed the impact of each feature in the choice of the class through beeswarms, barplots and violinplots. This analysis extends the information extracted by the correlation matrix and it displays how different features have different impacts in the prediction of different classes.

We then studied the model's fairness by investigating whether age and gender impacted the model's prediction differently (due to over/under-representativeness), displaying any bias toward a specific category toward a particular combination of age and gender. First, we grouped the population into three age ranges: young (<30 y.o.), middle-aged (31-50 y.o.) and old (>50 y.o.). Then, we produced 3 boxplots:

- one showing the overall distribution of SHAP impact magnitude by age group,
- one showing the overall distribution of SHAP impact magnitude by gender,
- one showing the overall distribution of SHAP impact magnitude by each combination of age and gender.

Our findings will be discussed later in this document.

Robustness and external validation analyses are deemed unnecessary due to the tabular nature of the data and due to the characteristics of the classifier itself. We would provide the final user with a questionnaire having predefined numerical ranges to be chosen; this would guarantee total adherence with the training data and it would avoid issues regarding possible outliers and unattended values.

Navigating through the 'app IO', the user will encounter a section dedicated to the fill of the questionnaire. Their answers would be collected and sent to a centralized server which will actually estimate the class prediction and return the suggestion to the user. The adoption of the 'app IO' guarantees both the quality of the data (their provenance from actual individuals) and the age of the users (since the access is provided to adults only); these are additional information which might be crucial for future research which might be aimed at segmenting and clustering the population and to drive micro-targeted awareness campaigns for the prevention of gambling addiction.

General safety and performance requirements

BetWise software relies its predictions on patient-provided data, collected through a structured questionnaire, to make its classification assessments regarding the risk of gambling addiction. Since this data is personal and sensitive in nature, particularly within the context of health, it is crucial that its handling complies fully with the General Data Protection Regulation (GDPR) for EU citizens.
To ensure compliance, we require the explicit consent of the user before any data is collected or processed. This consent is requested during the initial launch of the plugin on the user's smartphone.

The following steps have been implemented to meet safety and performance standards and GDPR obligations:

- Transparent communication: clear and concise privacy notices are provided to all users, explaining how their data will be used and stored.
- Purpose limitation: the collected data will be used exclusively to enable and enhance the predictive functionalities of the medical device. Any additional use for health-related research in other fields will require explicit user consent, as outlined in the post-market monitoring plan.
- Data minimization: only the data essential for accurate risk prediction is collected. No unnecessary or unrelated data points are included.
- Data quality: regular review and updating of the data used in the AI model to maintain high accuracy and relevance levels.
- Data retention and anonymization: once data is no longer needed, it is either anonymized or securely deleted.
- Security measures: encryption, access control, and other technical safeguards are implemented to protect personal data from unauthorized access or breaches.
- Human oversight: the medical device includes a human-in-the-loop interaction model. This ensures that no critical decisions affecting patient safety are made solely by the system, thereby mitigating risk and preserving clinical oversight.

By incorporating these mechanisms, our plugin adheres to essential safety and performance requirements, ensuring ethical, secure, and responsible use of patient data throughout its operation.

AI performance requirements

The AI system demonstrates robust predictive performance with an overall F1-score of 0.93, balancing precision and recall to address class imbalances in its multi-class classification task. While the Matthews Correlation Coefficient (MCC) was considered for its reliability in imbalanced scenarios, the F1-score was prioritized for its easier interpretability and alignment with the device's role as a decision-support tool under human oversight.

Since all data is gathered through user-completed questionnaires with predefined, bounded response options, there is no possibility for outliers to enter the dataset (every response adheres to a finite, well-defined range). This built-in consistency guarantees that model predictions reflect authentic user behavior rather than anomalies.

Explainability analysis using SHAP values revealed some fairness considerations. Older individuals systematically receive higher feature importance scores, suggesting age-related bias. Gender disparities were also observed, with male predictions exhibiting wider variance in SHAP impacts compared to females, raising concerns about indirect discrimination. Intersectional risks, such as compounded sensitivity for groups like 'Old Male' and 'Middle-aged Male,' further underscore the possible targeted bias. These findings emphasize that while outliers are absent, unintended outcomes can still arise from systemic biases in feature reliance.

To mitigate these challenges, the system adheres to GDPR, AI Act Article 14, and ALTAI principles through proactive human oversight, pre- and post-deployment bias audits, and continuous monitoring with tools like SHAP and validation curves.

Benefit and risk-analysis

A comprehensive benefit-risk analysis has been conducted in accordance with Sections 1, 3, and 8 of Annex I of the Medical Device Regulation (MDR), and is fully aligned with the risk management and quality assurance requirements of ISO 13485 and ISO 9001.

Given the tool's non-invasive nature and absence of physical interaction with users, safety-related risks are minimal. However, psychological and cognitive risks (such as over-reliance on predictions or misinterpretation of outputs) are recognized as key concerns. Identified hazards include algorithmic bias, model overfitting, and user misinterpretation. These risks are addressed through ISO 13485-compliant design controls, including fairness-aware model training and routine performance validation, as well as ISO 9001-driven continuous process improvements, such as periodic model retraining and integration of user feedback.

Risk mitigation is further supported by clear labeling strategies, including in-app warnings (e.g., "This tool does not replace professional diagnosis"), aimed at managing user expectations and promoting responsible use. Residual risks (such as potential delays in clinical diagnosis resulting from false negatives) are thoroughly documented and actively monitored through a post-market surveillance plan, ensuring ongoing compliance with ISO 13485 throughout the device lifecycle.

Clinical evaluation

As previously described, the reference literature on online gambling behavior is based on the studies such of **LaBrie et al.** and **Braverman et al**., who, using the same dataset, identified behavioral risk markers through an unsupervised clustering approach. In particular, Braverman et al. classified gamblers into four clusters, highlighting significant differences between groups in terms of frequency, intensity, and variability of betting behavior.

When comparing the results obtained by our supervised model with those reported by Braverman et al., some interesting parallels emerge. In our case, the most influential features in classification were the total number of stakes, the standard deviation of stakes during the first month, the average number of daily bets, and the total number of active days. SHAP analysis showed that *p2net* (net expenditure from live action bets: stakes minus winnings) played a major role in all clusters, while *gender* (user's binary gender: male or female) and *Zp2totalactivedays1m* (z-score of total active days in live action during the first month) were particularly relevant for clusters 2 and 4.

In the study by Braverman et al., the four identified clusters exhibited statistically significant differences across key behavioral metrics (frequency, intensity, variability, and trajectory), with F values ranging from 61.6 to 149.8 ($P < 0.001$).
The high-intensity, high-variability cluster (cluster 1), in particular, was associated with a high rate of account closures due to gambling-related problems (73%; $\chi^2 = 13.58$, $P < 0.01$), and higher monetary losses ($P < 0.05$). However, no single variable was found to significantly predict this outcome. Compared to these findings, our supervised approach allows for a more targeted modeling of the impact of individual features on each class, offering a more granular interpretation of risk-related behavioral dynamics.

Post-market surveillance

Our AI-based gambling risk assessment tool will undergo continuous post-market monitoring to ensure its reliability, fairness, and alignment with real-world user needs. The App IO platform team will manage day-to-day monitoring, leveraging the existing infrastructure for data collection, user feedback channels, and system performance tracking. Our team will collaborate through specialized AI evaluations, conducting model accuracy checks with a focus on underrepresented high-risk user groups, fairness audits across age and gender demographics, and statistical analyses to detect shifts in user behavior (e.g., changes in betting frequency or variability).

Quarterly reviews will evaluate performance trends, user adherence to medical consultation recommendations, and any emergent biases. If significant data drift or performance decay is identified, we have built in the option to propose a model retraining; however, recognizing that we do not receive a clinical diagnosis for every model prediction, such retraining may only proceed if collaborating healthcare professionals provide diagnostic feedback to confirm label correctness.

Future development

Additionally, future developments may extend the academic relevance of BetWise by allowing researcher and policymakers to extract more information from the collected data which could support targeted advertising and awareness campaigns, driving a positive impact on the challenge of gambling-related harm. We aim that future researchers could find our tool useful for population-level clustering analyses and segmentation.